

What is the relationship between the percentage of those covered by health insurance and median rent costs per counties in Texas in 2024?

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Summary:

The report examines the relationship between health coverage and median rental costs for counties in Texas. Using data sets pulled from American Community Survey, I produced analytical data in visual formats to describe the impact of rental costs in the rates of those who have health insurance. Lastly, I used an OLS regression that best describes the estimates of the impact.

Key Findings: A one dollar increase to the median rent causes the rates of coverage to increase 0.0046%. There is positive correlation between these two variables, but they are very small. It would be recommended to add on more potential variables such as “Average Travel Times” or “Walkability” for further analysis. Many counties in Texas vary greatly in population density and having variables that account for this is recommended.

Research Question and Motivation:

Can renting costs help in explaining the rates of those insured in a given county in Texas?

The motivation for the research is to find a correlation between the two variables and to help explain why some places see higher rates of an insured population than others. Understanding the relationship could be used investigate the economic performance of counties. Any correlation found in this study does not imply causation.

Data:

The data used in the analysis was pulled from the ACS API (American Community Survey) for the year 2024.

Variable	ACS Series	Description
SELECTED CHARACTERISTICS OF HEALTH INSURANCE COVERAGE IN THE UNITED STATES (%)	S2701_C03_001E	This is the ratio between insured and population for the United States.
MEDIAN GROSS RENT (DOLLARS)	B25064_001E	Gross rental costs for housing in the United States. Estimated values of utilities and fuel are also added into the calculation.

Sample Period: Pulled data from over a 5-year period but specified to only have the year 2024 when analyzing.

Data Quality Notes: No issues with health insurance rate data as there were no negative values. Median gross rent did have negative values for 5 counties but were flagged and read as missing values. So, the analysis used 249 observations instead of all 254 counties in Texas.

Methodology:

The estimated OLS model:

$$\text{Predicted Health Insurance Rate} = 78.3116 + 0.0046 (\text{Median Rent})$$

I used an OLS model to analyze how the cost of rent performs when it is used to estimate the rate of those insured. I chose the median gross rent as my independent variable as a measure of the cost of living in a county. I recognize that it's not the only factor but wanted to see how it performs in the analysis.

Results (Exploratory Findings):

When the county variables are presented on a distribution chart, they both present a bell shape. The insured rate (Fig. 1) shows a slight skew to the left indicating that those insured sit around 80-85% for a majority of counties. The median rent (Fig. 2) presents a more noticeable skew to the right as a majority of counties have their costs just under a thousand dollars.

The correlation matrix shows a result of 0.2581 which shows a positive correlation between the two variables. The findings from the scatter plot (Fig. 3) gives a better visual on how this positive correlation looks like. The blue line represents the linear regression of the data sets. The strength of the correlation is slightly low as it shows data being sparse in some areas. This would be due to the varying combinations of rent and insurance rates. However, there is still a portion that shows packing vertically and continues as rental costs increase.

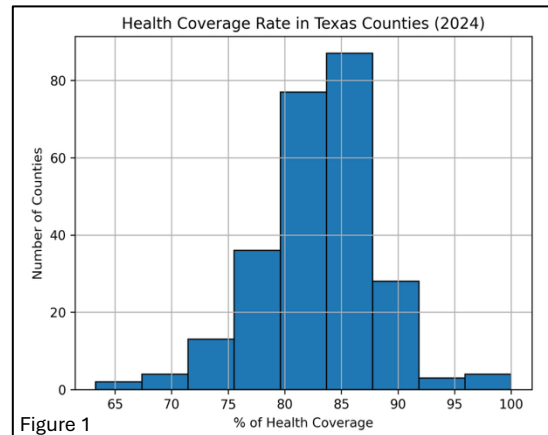


Figure 1

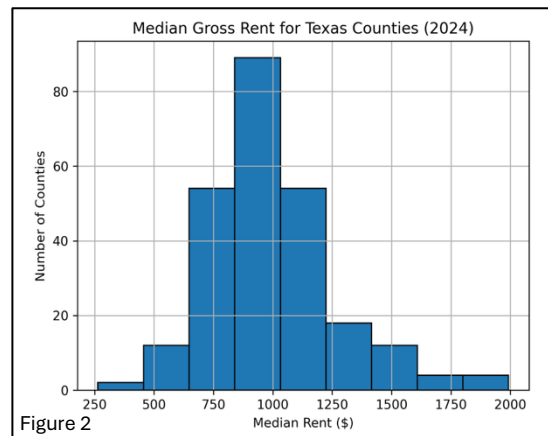


Figure 2

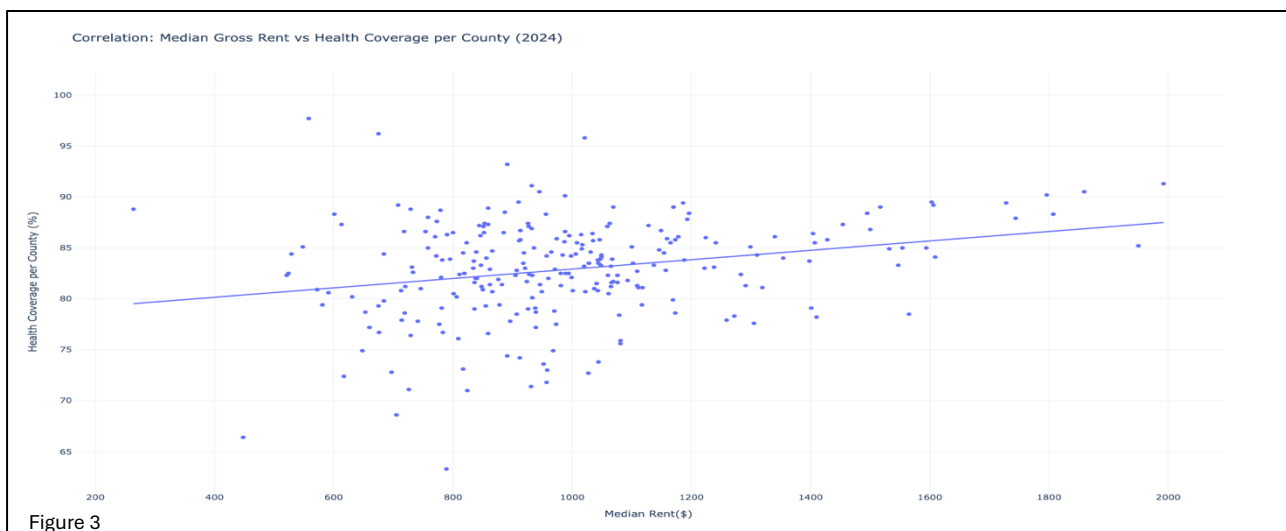


Figure 3

Results (Regression):

Variable	Expected Sign	Estimated Coefficient	Interpretation
Intercept (b0)	-	= 78.3116	If Median Gross Rent is 0, the predicted Health Insurance Rate would be 78.3116.
Median Gross Rent (b1)	Positive	= +0.0046	For every one dollar increase in the median rent, the Health Insurance Rate would increase by 0.0046.

R-Squared: The proportion of the variability in Health Insurance Rates that is explained using Median Rent. Which is 0.067, so 6.7% of the variation of Health rates can be explained with Rent costs.

Assessment: The results from the regression indicate that there is a slight increase in the percentage of those insured when the median rent cost increases. When analyzing the P-Values to a 95% confidence interval, it concludes that there is a potential use for rent costs to be an independent variable for the rate of insured. The interval is also positive and holds no value of 0 as it is between 0.002 and 0.007. This also concludes that the variable is potentially useful for the regression.

- Note that the regression output states potential numerical problems. See notebook “*balderas_midterm_executed.ipynb*” for more information.

Limitations:

- I recognize that there are limitations when discovering the relationship of the two variables. The first being that any correlation in the study does not imply causation was discovered. The increase of rental costs for a resident does not imply it caused a higher rate of the population to be insured.
- The second limitation is the samples that were omitted in the analysis portion due to it being unusable. There were negative values for certain counties in the median gross rent. Considering that it is in the unit of dollars, it wouldn't be possible. The sample size was decreased by 5 counties when I performed a left join merge of the two data sets.
- The third limitation is that this is a snapshot of data for the year 2024. There was no time trend analysis do see how values change over time together.
- Lastly, the units of the variables represent the entire population of those in a county. The analysis would struggle to represent population densities between rural and urban areas.

Implications for Practitioners:

- Business Analyst:** The data can better understand the relationship between higher costs of living and the rates of insured. It could also highlight counties that could be potential markets to insure the population.
- Policy Researcher:** The use of this analysis could better investigate certain counties economic performance and how rent prices could be a factor in the rate of those insured.

Conclusion:

The regression determines that using median rent as a variable could be potentially useful in explaining the rates of an insured population. However, only about 6.7% of the of the variation of insured rates are explained by the median rent. There are also the limitations of using precalculated units to represent entire counties with the data. I'd recommend adding more variables such as "walkability" or "travel times" into the analysis to account for population density. I would also recommend making it a time series analysis to see potential trends in the analysis.

AI Disclosure:

Google Gemini 3 Flash was used during the coding process of the project. The model was used to describe python code from past course lectures to better understand their implementation into my notebook. Codes were ingested into the AI model to clarify syntax error and troubleshoot problems with pulling Census data. I also asked it how to make a general distribution table in python code but set bin size and variables myself. Please see "*balderas_midterm_executed.ipynb*" for specific annotations on the use of AI for this project. **AI was not used in generating this practitioner report or took any part in the interpretation of the results.**

References:

- Penn State University Libraries Catalog
<https://catalog.libraries.psu.edu/catalog/47395275>

*This report was produced as an individual project for BANA 4373. All code can be replicated with the accompanied Jupyter Notebook "*balderas_midterm_clean.ipynb*".*